

ORCA

Marine Ecosystem Reasoning with Collaborative Agents

Team briefing document — SIH26176 | ISRO | Space Technology / Marine Ecosystem

1. Problem statement

Official reference: SIH26176, submitted by ISRO, theme Space Technology / Marine Ecosystem. India's marine ecosystem supports millions of livelihoods across fishing, coastal communities, and shipping, yet the satellite and oceanographic data that could protect and guide them—chlorophyll levels, sea temperature, tides, weather hazards—sits scattered across specialist portals in technical formats disconnected from each other. There is no accessible, intelligent system that reasons across these sources and responds in plain, trustworthy language, in the user's own language.

2. What we are building, in one paragraph

An agentic AI-powered conversational platform for marine intelligence. Users ask natural-language questions, such as whether it is safe to fish near a location tomorrow, and a team of specialised AI agents pulls real satellite and oceanographic data, reasons across it, and answers in plain language, in the user's own language, alongside an interactive map. The same engine serves fishermen, coastal patrol officers, maritime operators, coastal authorities, and researchers.

3. Novelty — what makes this different

- Collaborative multi-agent reasoning, not a single model or static dashboard.
- Conversational-first design; the map is a supporting visualisation, not the primary interface on its own.
- Automatic multilingual reasoning; detects the query's language and responds natively.
- Explainable verdicts with a confidence score and a cited evidence trail, never a black-box answer.
- One engine serving multiple stakeholder groups, reframed per audience rather than five separate tools.
- Built around India's ISRO and INCOIS data infrastructure.

4. Core features and outputs

Per location or region, the system produces:

- Chlorophyll level (mg/m³) and sea surface temperature, current reading.
- Algae bloom risk: low, medium, or high, compared to that location's historical baseline.
- Environmental fishing conditions: favourable, caution, or avoid.
- A 7–14 day historical trend chart.
- A 2–3 day forecast with a confidence band that widens further into the future.
- A confidence score (0 to 100) and plain-language evidence trail.
- Tide, wave, and wind conditions for that location.

5. SIH26176 capability coverage

The exact SIH26176 statement explicitly expects agentic planning, specialised-agent collaboration, multi-source marine/geospatial integration, spatial-temporal reasoning, multilingual and multi-turn conversation, explainability, proactive alerts, geofencing, route optimisation, and evidence-backed recommendations.

Official requirement	ORCA implementation	Status / priority
Natural-language intent	Planning Agent	Covered — P0
Regional-language response	Language detection + multilingual Reporting Agent	P0 — demonstrate 2–3 Indian languages
Multi-turn context	Session memory + conversational context	P0
Autonomous data discovery	Marine Data Discovery Agent + source/tool registry	P0
Spatial / temporal reasoning	Ocean Analytics + Geospatial Reasoning	P0
PFZ discovery	PFZ identification, ranking and map highlighting	P0 — strengthen current design
Hazard alerts	Weather Intelligence + alert service	P0
Geofencing	EEZ, MPAs, restricted/sensitive zones	P0
Route optimisation	Hazard-aware route cost + path selection	P0 — expand beyond safety check
Explainable recommendations	Risk Assessment + evidence/source panel	P0
Maps/charts/advisories	Leaflet + Recharts + advisory-style output	Covered — P0
Productivity decline reasoning	Historical anomaly + cross-source comparison	P1

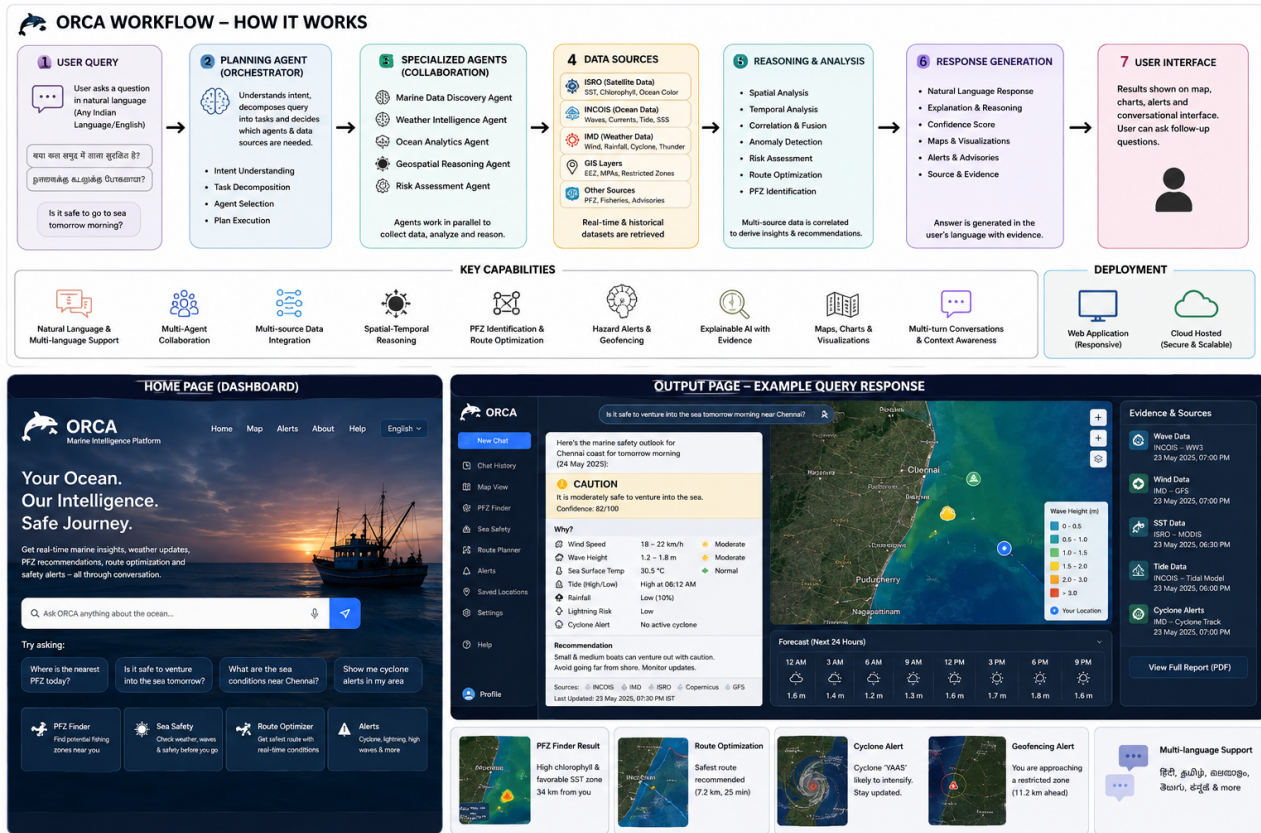
Important implementation interpretation: the project should remain a web application with a conversational-first interface and interactive map. It is not only a static analytics dashboard.

6. Additional outputs and operational features

- Short-range route safety check between two points, approximately 100–150 km and 1–2 day window.
- Geofencing alerts near India's EEZ boundary and marine protected areas.
- Proactive hazard alerts for cyclones, lightning, and high waves.
- Formatted advisory-style output similar to official INCOIS or IMD bulletins.
- PFZ finder: nearest suitable zone, distance, suitability factors, safety/geofence filtering.
- Follow-up questions that preserve location, activity, and context.

7. End-to-end workflow

Recommended product flow: the user asks in natural language; the Planning Agent resolves intent and context; specialised agents retrieve and analyse only the required data; a scientific/risk layer synthesises the evidence; the Reporting Agent returns the result with map, charts, alerts, confidence, and sources.



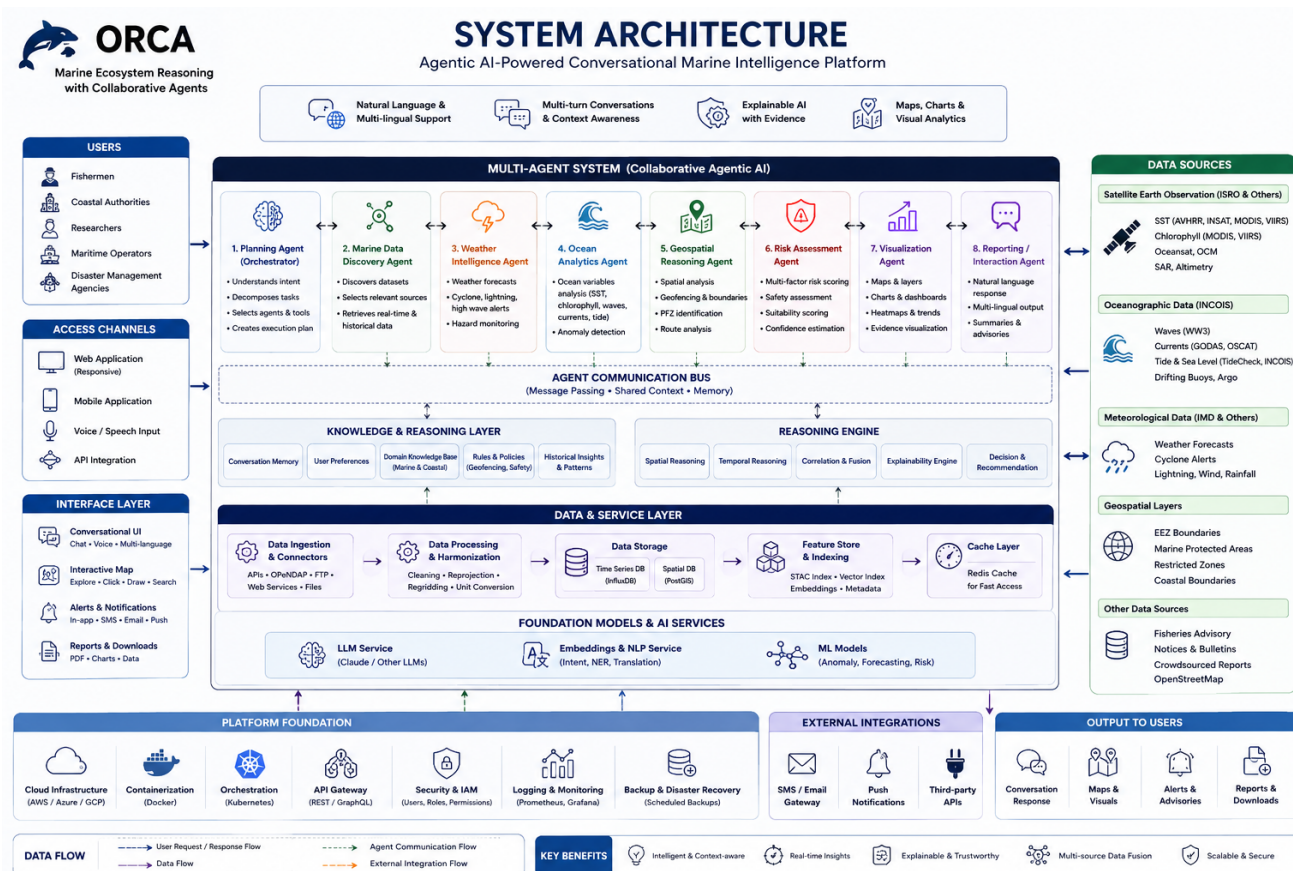
The workflow image is a design/implementation blueprint, not a screenshot of an already deployed system.

8. Agent responsibilities

Agent	Role
Planning Agent	Decomposes the user query into sub-tasks, resolves intent/time/activity, selects required agents and tools.
Marine Data Discovery Agent	Finds and retrieves the right satellite/ocean dataset for the query.
Weather Intelligence Agent	Cyclone, lightning, wind, wave conditions and proactive hazard alerts.
Ocean Analytics Agent	Chlorophyll, SST, bloom detection, anomaly analysis and forecasting.
Geospatial Reasoning Agent	Location resolution, EEZ/MPA geofencing, PFZ identification and route reasoning.
Risk Assessment Agent	Synthesises signals into a recommendation with a transparent confidence calculation.
Visualization Agent	Chooses map, chart, heatmap, route, PFZ or advisory presentation.
Reporting / User Interaction Agent	Conversational response, language detection, multi-turn context and user-facing explanation.

9. System architecture

The architecture follows the SIH-recommended modular multi-agent pattern. The central system coordinates specialised agents while a data/service layer normalises external sources and a reasoning layer performs spatial-temporal correlation. User outputs are delivered through the web application as conversation, maps, alerts and reports.



Implementation note: the diagram is the target architecture; individual services may be consolidated during the 36-hour MVP if necessary, without changing the product behaviour.

10. Proposed technology stack

Layer	Technology	Purpose
Frontend	React + Tailwind CSS	Responsive web application, conversational UI, dashboard views.
Map	Leaflet	Scientific/GIS map, overlays, markers, geofences and routes.
Charts	Recharts	Historical trends and forecast/confidence bands.
Backend	Python + FastAPI	Async APIs, data connectors, agent services and decision engines.
Agent orchestration	Plain Python functions / lightweight orchestration	Fast to implement and debug during hackathon.
LLM	Claude API or equivalent	Intent, orchestration, explanation and multilingual response.
Forecasting	pandas / numpy baseline; LSTM only if time permits	Start simple and working; improve only after MVP is stable.
Geospatial	xarray, rasterio, geopandas, shapely	Satellite/raster processing, spatial joins and geometry operations.

11. Implementation — core services

11.1 Web application

- Home page: conversational query box, quick actions, live conditions, active alerts and map preview.
- Chat mode: natural-language or voice input, multi-turn context, map auto-pan/highlight.
- Map Explorer: layers for chlorophyll, SST, waves, wind, PFZ, hazards, EEZ/MPA and routes.
- PFZ Finder: nearest zone, ranked alternatives, distance, productivity and safety evidence.
- Sea Safety: verdict, confidence, supporting variables, forecast, recommendations and advisories.
- Route Optimizer: origin/destination, candidate routes, hazards avoided, distance and estimated time.
- Alerts: proactive hazard notifications and geofence warnings.
- Reports: advisory-style summary and exportable report.

11.2 Data ingestion and normalisation

- Create one adapter per source with a common internal interface.
- Normalise timestamps, units, coordinates, missing values and quality flags.
- Cache frequently requested datasets/results for hackathon reliability.
- Display source name and data timestamp with every important recommendation.

11.3 Scientific decision layer

- **Bloom risk:** current chlorophyll versus local historical baseline + INCOIS bloom index.
- **Environmental fishing conditions:** productivity + SST + sea-state suitability; explicitly not a catch guarantee.
- **Safety:** waves + wind + tide + rainfall + lightning + cyclone/advisory conditions.
- **PFZ:** candidate-zone generation and ranking using productivity and safety/geofence filters.
- **Route:** hazard/geofence-aware cost surface and lowest-cost route.
- **Confidence:** derived from data freshness, source availability, forecast horizon, spatial uncertainty and signal agreement; not invented by the LLM.

11.4 LLM boundary

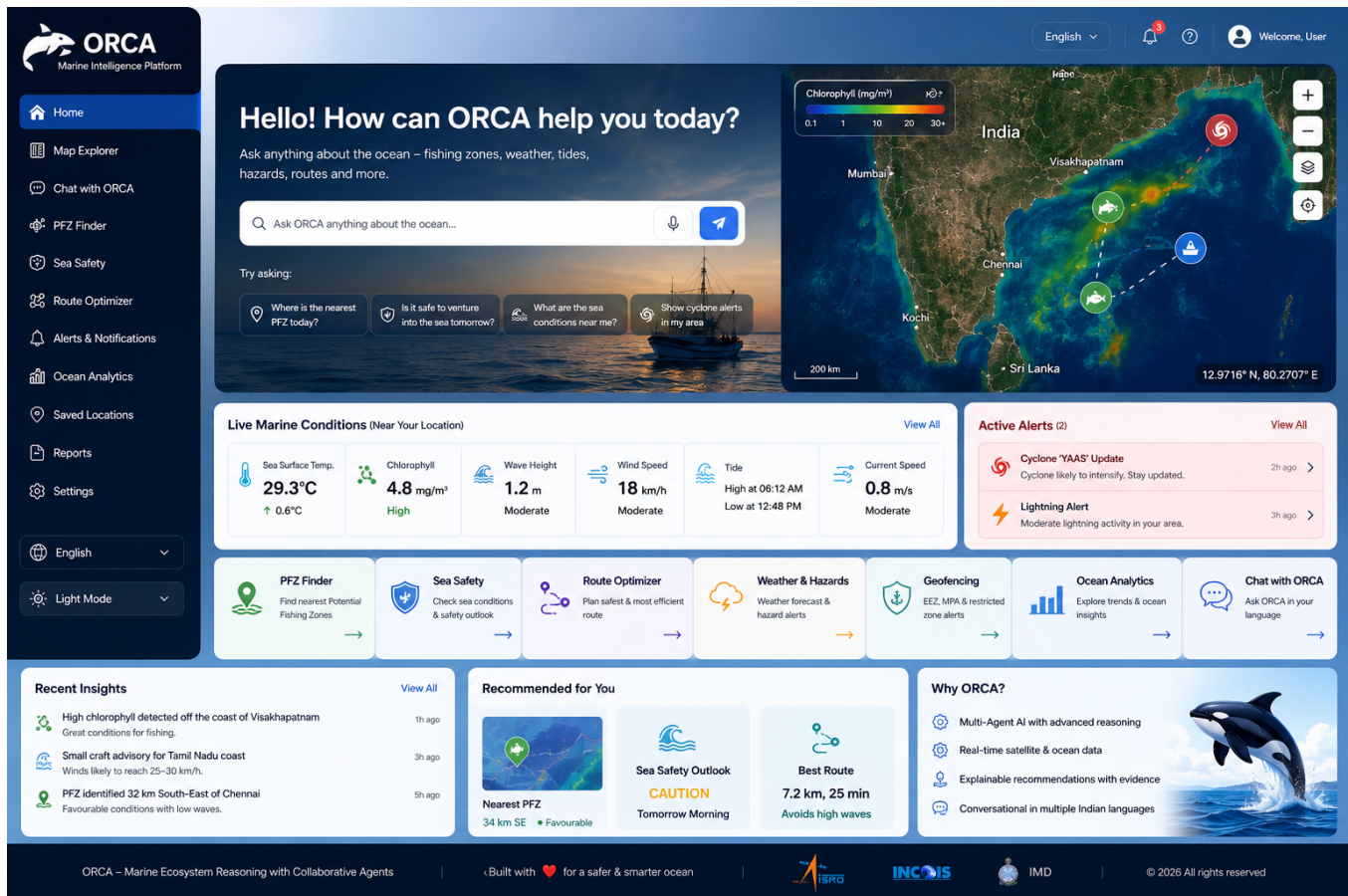
The LLM should interpret intent, select/coordinate tools, preserve conversational context, and explain structured results. It should not invent ocean measurements, safety scores, PFZ locations, or confidence values. Scientific calculations should be generated by the data/analytics layer and passed to the LLM as structured evidence.

12. Data model and reliability

Canonical field	Example
timestamp	2026-08-23T18:00:00+05:30
location	lat/lon or geometry
variable	sst, chlorophyll, wave_height, wind_speed
value / unit	30.2 / °C
source	INCOIS, MODIS/VIIRS, IMD, etc.
quality / freshness	quality flag + retrieval time

13. Web application — home page

The proposed product should be implemented as a responsive web application. The dashboard is the shell around the experience; the conversational query is the primary interaction, while the map is a supporting visualisation driven by the same underlying engine.

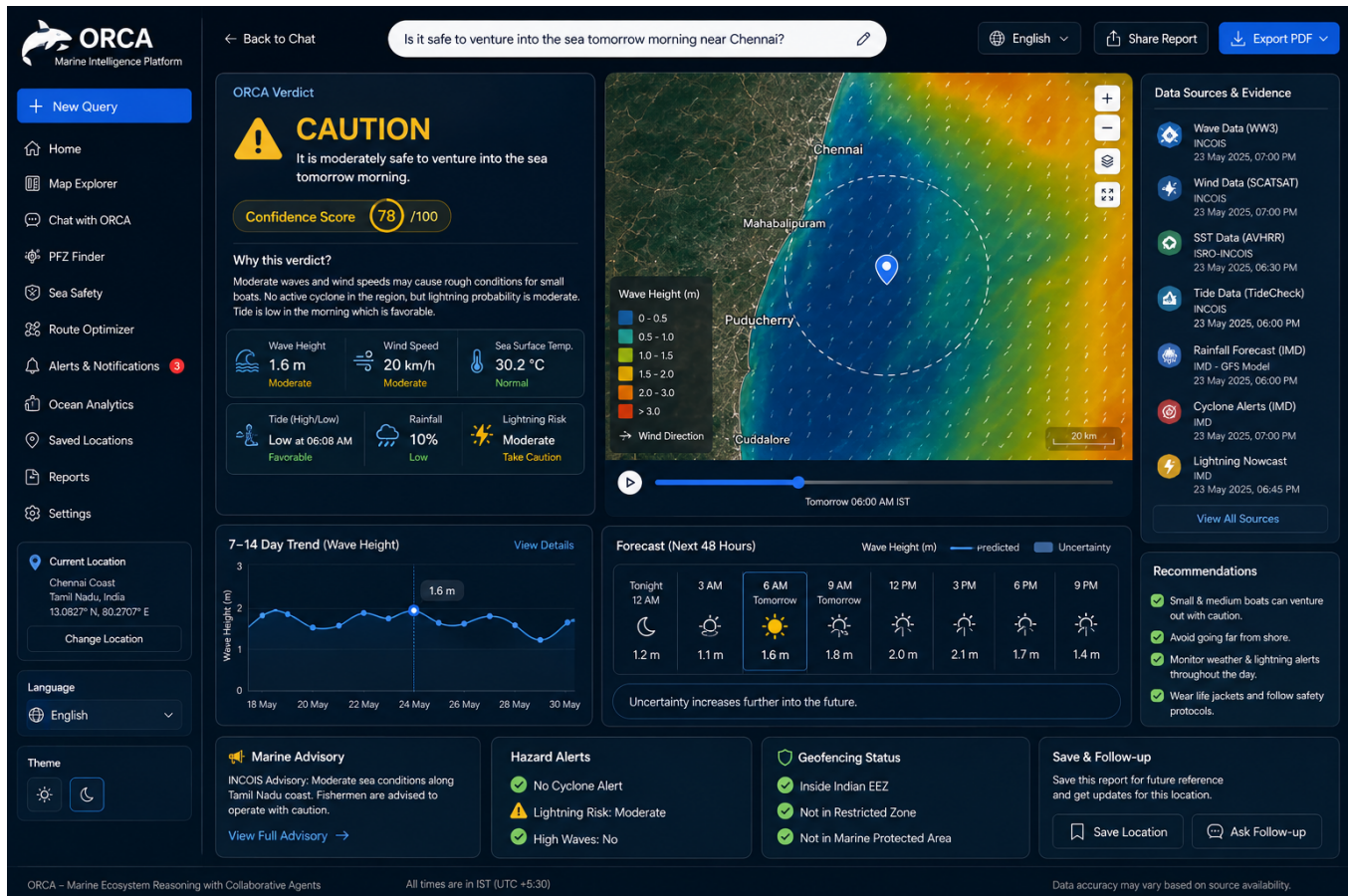


Home-page design priorities

- One prominent query box: ask about PFZs, safety, tides, weather, hazards, routes and ocean conditions.
- Quick actions for PFZ Finder, Sea Safety, Route Optimizer, Weather/Hazards and Geofencing.
- Live marine conditions and active alerts should be visible without requiring a query.
- Map preview should show meaningful layers but not overwhelm the conversational experience.
- Regional-language selector and voice input can sit at the interaction layer.

14. Web application — example output

A good output page should answer the user's question immediately, then expose the supporting evidence. The design below shows a safety query; the same layout can be adapted for PFZ, route, productivity, or alert queries.



Recommended output structure

- **Verdict:** prominent SAFE / CAUTION / AVOID-style advisory.
- **Confidence:** numeric score with a short explanation of its basis.
- **Why:** the key environmental/hazard signals driving the recommendation.
- **Map:** exact relevant zone, route, hazard overlay or PFZ.
- **Forecast:** short-range time series with widening uncertainty.
- **Evidence:** source, timestamp and relevant observation.
- **Recommendations:** practical next steps, with limitations clearly stated.

15. What is still open — action items before the hackathon

- Assign team roles: frontend (map/chat/UI), backend (FastAPI/agents), data pipeline (ISRO/INCOIS downloads).
- Register on MOSDAC; full access needs signup and email confirmation.
- Confirm IMD cyclone and lightning data access.
- Build one complete end-to-end live query before adding stretch features.
- Implement PFZ discovery and demonstrate it on the map.
- Upgrade route safety into actual hazard-aware route optimisation.
- Implement 2–3 Indian regional-language demonstrations.
- Implement real multi-turn follow-up context.
- Create proactive hazard/geofence alert workflow.
- Validate bloom detection against a known historical Indian coastal bloom event.
- Prepare 2–3 live demo questions plus a recorded backup video.
- Prepare the final pitch deck using this briefing as the technical/content backbone.

16. Scope discipline — what we are intentionally not building

- Long-haul, multi-day voyage route planning; keep the MVP to a justified short-range forecast window.
- Fish species identification; chlorophyll indicates general productivity, not species presence.
- Vessel tracking or surveillance capability for defence use; ORCA is an environmental and geofencing awareness layer, not a tracking system.
- Complex LSTM forecasting before the core real-data pipeline is stable.

17. Recommended SIH demonstration flow

Demo	User asks	What judges see
1 — PFZ	“Where is the nearest Potential Fishing Zone today?”	Planning → ocean data → PFZ ranking → geofence/safety filtering → map + evidence.
2 — Safety	“Is it safe to venture into the sea tomorrow morning?”	Weather + waves + tide + hazards → risk verdict + confidence + advisory.
3 — Route	“Find the safest route to this fishing zone.”	Route alternatives → hazard/geofence cost → recommended route + reasons.
4 — Follow-up	“What about the afternoon?”	Same location/context preserved; only requested time dimension changes.

ORCA should be built as a conversational marine decision-support web application, not as a static dashboard. The strongest SIH story is: a user asks a real marine question; ORCA autonomously plans the task, selects relevant data and agents, correlates multiple observations, performs spatial-temporal reasoning, and returns an evidence-backed recommendation through conversation, maps, alerts and charts.

The generated UI and architecture images in this document are design references for implementation. They should be treated as target layouts, not claims of current functionality. The original team briefing remains the source for the current verified data-source and scope statements.